Gagan Sharma

Bengaluru, Karnataka
+91 7006387398 gagansharma11564@gmail.com GitHub LinkedIn Portfolio

Professional Summary

Data Scientist focused on **Machine Learning** and **MLOps**, building end-to-end systems from model development to deployment. Proficient in **Python**, **SQL**, Scikit-learn, and XGBoost, with hands-on experience in **feature engineering**, model evaluation, and statistical analysis. Built production-style projects spanning **FastAPI**, Docker, CI/CD, model monitoring, and **Retrieval-Augmented Generation (RAG)** with LLM-based document retrieval and generation.

Skills & Technologies

Programming: Python, SQL
Machine Learning: Scikit-learn, XGBoost, Classification, Regression, Feature Engineering, Hyperparameter Tuning, Model Evaluation, Recommendation Systems
Deep Learning & AI: TensorFlow, Keras, CNNs, Computer Vision, RAG, LLMs, Semantic Retrieval, Cross-Encoder Reranking
MLOps & Deployment: MLflow, FastAPI, Docker, Git, GitHub Actions, DVC, CI/CD, Model Monitoring, Canary Deployment, Model Rollback
Data Analysis: Pandas, NumPy, EDA, Statistical Analysis, Data Cleaning & Preprocessing
Databases & BI: MySQL, PostgreSQL, MongoDB, ChromaDB, Power BI, Tableau, Excel

Projects

Payment Intelligence & Smart Routing Platform | GitHub | Live Demo | API Docs

- Simulated **100K+ transactions** across 5 payment methods; failure rate was **33.0% on poor networks vs. 14.7% on excellent**.
- Trained Logistic Regression and Random Forest failure-prediction models (**Scikit-learn**) with engineered features, probability calibration and risk-bucket analysis.
- Built a **risk-aware routing engine**; in simulation it lowered expected failure risk from **17.4% to 15.4%**. Served with **FastAPI, PostgreSQL, Docker**.

Enterprise Document Intelligence & RAG Assistant | GitHub | Live App

- Built and deployed a **RAG application** for enterprise document Q&A using **FastAPI, Streamlit, ChromaDB**, Sentence Transformers, and cross-encoder reranking, with citations and **Groq/Gemini fallback**.
- Evaluated retrieval on a **40-question benchmark** over 8 policy documents: **100% Hit Rate@5**, with reranking raising **MRR from 0.975 to 1.000** at **108 ms** mean retrieval latency (vs. 16 ms without).
- Implemented **Supabase authentication** with password recovery, **Redis-backed sessions** (opaque IDs, 7-day TTL), and per-user workspace isolation; shipped with Docker, GitHub Actions CI, and **130 automated tests**.

Customer Churn MLOps System | GitHub | Live API

- Built a churn prediction system on 7,043 records using **XGBoost** and GridSearchCV (**ROC-AUC 0.846**, **72% recall**); tracked in MLflow, served via **FastAPI** and **Docker**.
- Automated **CI/CD** with GitHub Actions, a ROC-AUC ≥ 0.80 quality gate, and **Evidently drift monitoring** (flags at 50%+ columns drifted).
- Designed a **90/10 canary deployment** on Render that promotes only if F1 ≥ 0.50, ROC-AUC ≥ 0.75 and no worse than production; failures trigger **automated rollback**.

Experience

Analyst Trainee – Savantis Solutions India Pvt. Ltd. **Dec 2025 – Jul 2026**
Pfizer Project Team, HCL Technologies *Hyderabad, India*
Pre-assignment training: Oct–Dec 2025

- Built **SQL queries** and **Power BI dashboards** automating Excel reporting, cutting team data-handling time **22%** (manager-verified).
- Owned **end-to-end SQL reporting**: queries, report generation, **data validation**, testing, and QA, with team-lead sign-off.
- Used **SQL validation** and **root-cause analysis** to catch data inconsistencies and resolve access discrepancies in **ServiceNow**.

Tester Intern – Wyreflow Technologies **Oct 2024 – Nov 2024**
Software Testing Team *Bangalore, India*

- Performed **functional and regression testing** on web applications, executing **50+ test cases**, validating **APIs**, and documenting defects for resolution.

Data Scientist Intern – Learnbay **Jan 2024 – Sept 2024**
Data Science Team *Bangalore, India*

- Performed **EDA**, preprocessing, and feature engineering using **Python, Pandas, and NumPy**.
- Developed **ML models** and **CNN-based image classification** using **Scikit-learn, TensorFlow, and Keras**, achieving **92% accuracy**.

Education

SDM College of Engineering And Technology **Dharwad, Karnataka**
Bachelor of Engineering – Civil Engineering | CGPA: 8.05/10 *2023*

Certifications

Data Analysis – Cisco (2024) [Verify] **Python Programming – Udemy (2024)** [Verify] **Data Science (2024)** [Verify]